

Advancing Phase Change Materials (PCM) Technology: Research, Development, and Optimization

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Short Report

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Abstract

Phase Change Materials (PCMs) play a pivotal role in various heating and refrigeration applications by leveraging the latent heat exchange during phase transitions. However, recent demands for PCMs with higher melting temperatures, superior mechanical properties, enhanced thermal conductivity, and reduced reactivity to environmental conditions have necessitated a focused research effort.

Our team embarked on a systematic exploration of alloy compositions theoretically suited to meet these stringent requirements. Employing advanced techniques in material science, we conducted comprehensive analyses to identify promising candidates with the desired attributes.

Subsequent experimental phases involved the fabrication and thorough testing of these alloys under controlled conditions. Through meticulous evaluation of their performance characteristics, including thermal behavior, mechanical strength, and environmental stability, we aimed to pinpoint the most suitable material for PCM applications.

Our research endeavors not only contribute to the advancement of PCM technology but also hold significant implications for energy efficiency and sustainability in diverse industrial sectors.

Introduction

Phase change materials (PCMs) are important for their exceptional ability to release and absorb significant amounts of latent heat, rendering them indispensable for thermal buffering and energy storage applications. With prominent uses in heating, refrigeration, and solar panels, PCMs have been pivotal since the early 19th century.

Organic and inorganic compounds constitute the two main categories of PCMs. Organic PCMs, further classified into paraffin and non-paraffin types, primarily feature low melting points, making them suitable for low-temperature applications, with paraffins being the most prevalent. On the other hand, inorganic PCMs encompass salt hydrates and metal alloys. While salt hydrates exhibit reactivity at elevated temperatures and possess suboptimal strength and conductivity, metal alloys emerge as promising candidates due to their high melting points, exceeding 500°C, earning them the designation of high-temperature PCMs.

Metal alloys, a subset of inorganic PCMs, have garnered attention for their potential applications in solar panels over the past two decades. Despite undergoing extensive research, their utilization has yielded mixed outcomes [1-3]. Important considerations for their suitability in solar panel applications include durability over multiple heating cycles, minimal precipitate formation during heating, or controlled precipitation that does not escalate over time, and negligible volume variation during phase transitions.

In light of these requirements, thorough investigation and optimization of metal alloy PCMs are imperative to unlock their full potential for enhanced performance and reliability in solar energy systems.

Experiment and Results

Our goal with this research was to find metal alloy PCM with a melting point around 700°C and 900°C, a density greater than 3 grams/cm³ but less than 9 grams/cm³, and low flammability.

The project started with the identification of candidate metal alloys or compounds from the scientific literature. We also looked for correlations between physical and chemical properties of certain mixtures and their behavior with heating. Through a comprehensive literature review we found that some theoretical studies were performed on alloys like PCM, but there was insufficient testing [1-3]. We concluded that the current body of research on PCMs is relatively small with little practical testing. Based on previous research, we chose the alloys we wanted to construct and test

The alloy mixtures we chose were:

- 37% Si and 63% Ni mixture by weight,
- 56% Mg and 44% Si mixture by weight,
- 56% Cu, 27% Si, and 17% Mg mixture by weight, and
- 64.2% Fe, 26.38% Si, and 9.35% B mixture by weight.

These alloys fit our constraints of temperature, density, and flammability.

Due to time constraints in procuring materials, we were only able to construct and test the candidate mixture containing 64.2% Fe, 26.38% Si, and 9.35% B by weight (Fe_{64.2}Si_{26.38}B_{9.35}). This was the proposed eutectic mixture of those 3 metals. We predicted that this eutectic mixture would be the most thermally stable due to its exact melting point (rather than a range of melting points). We decided to construct 2 versions of this alloy: one containing Fe-26.38Si-9.35B by weight (Alloy A, the original proposal) and another containing Fe-29Si-10B by weight (Alloy B). Alloy A is the center eutectic point and alloy B is on the edge for the eutectic range.

We proposed two methods to construct these alloys. The first method used pressure to combine the metals. We grinded each metal component into a sand-like powder. Each metal powder was then weighed separately to get the correct proportions. Once the correct amounts were weighed, they were combined and mixed thoroughly. They were then put into a metal cylinder and compressed with hydraulic press to produce a pill-like object. During testing, this method was shown to be ineffective and unsuitable for further use.

The second method combined the metals using heat. Instead of using a powder, whole metal pieces were used. The pieces were cut to a size that yielded the appropriate weight. For Fe_{64.2}Si_{26.38}B_{9.35} we used 3 types of metals alloys we already had access to: an iron boron mixture, an iron silicon mixture, and pure iron. To produce Fe_{64.2}Si_{26.38}B_{9.35} the proportions of each mixture was 48.6% FeB, 34% FeSi, and 17.4% Fe by weight. These cut and weighed metals were placed in a crucible, which were then placed in a

furnace. The alloys were heated to 1650 °C, remained at 1650 °C for two hours, and were then cooled to room temperature. This was repeated three more times, heating to 1650 °C for 40 minutes, and then cooling to 1050 °C. After these four cycles, two small pieces were cut off each alloy for analysis. Then the rest of the alloy was put back in the furnace and cycled from 1650 °C to 1050 °C (staying at 1650 °C for at least 40 minutes) six more times. After the six cycles (10 cycles total) were done, two pieces were taken off each alloy for analysis.

One piece from the 4-cycle alloy and one piece from the total 10 cycle alloy were sent to another lab to test for conductivity, hardness, and strength and imaged through an electron microscope. The other 2 pieces from each alloy were kept for microstructure analysis. The pieces were prepared to view their microstructures. They were first wrapped into Bakelite, in a cylinder shape, where only a small piece of the metal was visible on one of the ends. The side with the metal visible was sanded down starting at 100 grit and ending at 1000 grit. It was then polished with an Al₂O₃ and water solution, as shown in Figure 1.

We then took images of the polished metal under a microscope going from 50 zoom to 1000 zoom. The alloy containing Fe-26.38Si-9.35B (Alloy A) and the alloy containing Fe-29Si-10B (Alloy B) both had the correct characteristics of having a melting point above 700°C and their densities were between 3 and 9 grams per cubic centimeter. Both alloys lasted multiple cycles with no indication of deterioration. In addition, they did not produce any precipitates and had minimal volume change when going through the phase change. So, we accomplished one of our goals.

We also wanted to see if there were changes from 4 cycles to 10 cycles. We found that there were no changes for these alloys.

Lastly, we wanted to find a eutectic microstructure in the alloys, to investigate the possibility of creating an ideal PCM and to analyze the difficulty expediency of producing it. Figures 2 and 3 below show some of the many images taken.

The little scribble-like structures in the images above are eutectic microstructures. Alloy A, (Figure 3) had more eutectic microstructures that were also more prominent, than Alloy B (Figure 2). We predicted that the tested alloy ratios would create a eutectic material based on research. Typically, there is only one binary eutectic ratio, but because there are three different metals, there is a range. We chose the nominal chemical composition (Alloy A) and the range's maximum (Alloy B). Alloy A seemed to have more prominent eutectic structures than B. We hypothesize that alloy B may be less prominent because it is at the end of the range, while alloy A is the global constant, but more trials are needed to confirm this. Because both alloys had these eutectic structures, we know they had a high latent heat of fusion. These eutectic structures also indicate that there is an exact melting point instead of a melting point range. This implies the reactions at different temperatures are predictable, therefore we could anticipate which alloy will be more thermally stable with varying temperatures.

Discussion

Phase change materials (PCMs) are substances renowned for their ability to absorb or release heat during phase transitions, known as latent heat. This unique characteristic makes them invaluable for thermal energy storage (TES) applications. While low-temperature PCMs have been extensively researched since the late 19th century, their limited melting points and poor conductivity render them unsuitable for high-temperature environments.

Recently, there has been a surge of interest in metal alloys as high-temperature PCM candidates due to their elevated melting points, superior conductivity, and high volumetric energy densities. However, the practicality of metal alloys for TES applications has been hindered by their degradation after numerous heating cycles.

To overcome these challenges, our research endeavors focused on identifying optimal metal alloy PCM candidates. Initially, we meticulously selected suitable metal alloys based on their properties. Subsequently, we prioritized investigating eutectic metal alloys due to their advantageous characteristics.

Our experimental approach encompassed the fabrication and comprehensive testing of PCM samples, including microstructure imaging and analysis of latent heat, conductivity, and melting points. Notably, we observed that specific alloy ratios resulted in eutectic structures, offering precise melting points, thermal stability, and exceptional heat storage capacities per unit volume.

These significant findings contribute to the ongoing quest for durable high-temperature metal alloys suitable for TES applications, ultimately advancing the efficacy and sustainability of thermal energy storage technologies.

References

1. Atinafu, Dimberu G., et al. "Thermal properties of composite organic phase change materials (PCMs): A critical review on their engineering chemistry." *Applied thermal engineering* 181 (2020): 115960.
2. Huang, Jintao, et al. "Advances and applications of phase change materials (PCMs) and PCMs-based technologies." *ES Materials & Manufacturing* 13 (2021): 23-39.
3. Wang, Xiaonan, et al. "A critical review on phase change materials (PCM) for sustainable and energy efficient building: Design, characteristic, performance and application." *Energy and Buildings* 260 (2022): 111923.

Figures



Figure 1

Polished metal in Bakelite

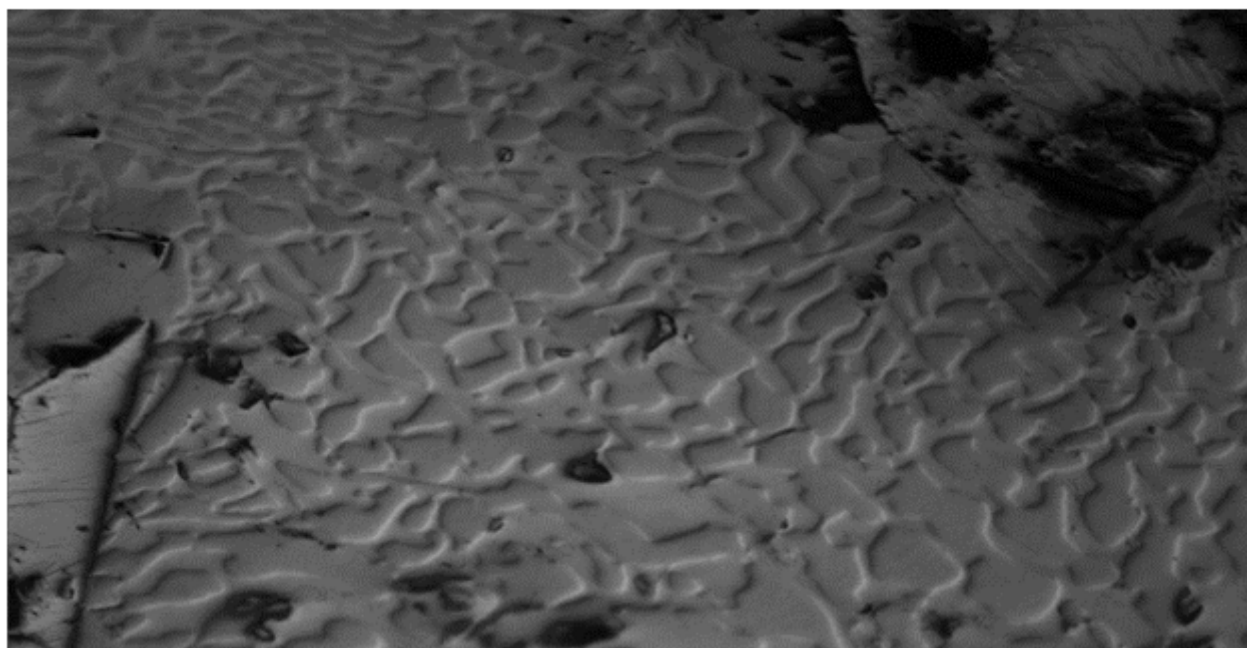


Figure 2

Alloy B, 500x zoom, 10 cycles

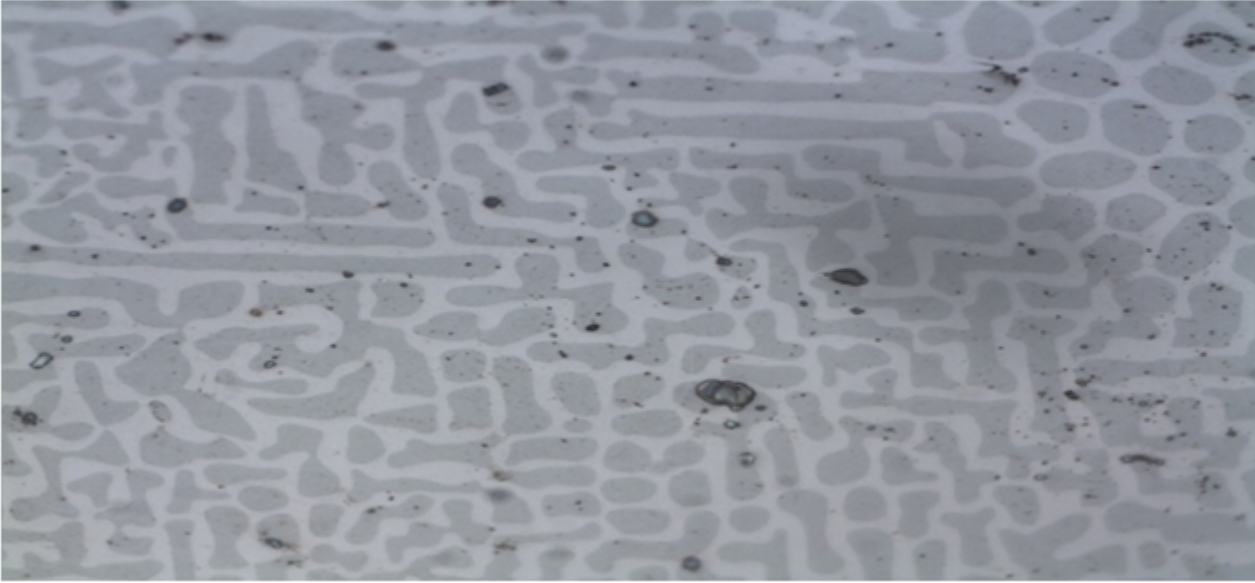


Figure 3

Alloy A, 1000x Zoom, 10 Cycles